📚 Modules 1-3 Checkpoint Exam Notes 🎯

🌐 Network Basics & Communication

- **Peer-to-Peer** 🤝: A network where a computer acts as **both client and server**.
- **Open Standards** 🌍: Allow devices with **different OS/vendors** to communicate and exchange data successfully (encourages innovation & competition).
- **3 Elements of Communication** 📡: Source (Sender) 📤, Destination (Receiver) 📥, and Channel 🔗.
- **Delivery Options** 📬:
  - **Unicast**: 1-to-1 👤
  - **Multicast**: 1-to-a specific group 👥
  - **Broadcast**: 1-to-ALL devices 📢

📏 Rules, Protocols & Models

- **Flow Control** 🚦: Prevents a sender from overwhelming a receiver (stops dropped packets).
- **Response Timeout** ⏱️: How a device reacts when a server takes too long to reply.
- **Layered Models Benefits** 🧱: Prevents changes in one layer from affecting others & assists in protocol design.
- **Protocol Stack Decoding** 📥: When a client *receives* data, it de-encapsulates **bottom-up** (e.g., **Ethernet ➡️ IP ➡️ TCP ➡️ HTTP**).
- **Layer Mapping** 🔄:
  - OSI **Data Link** + **Physical** layers = TCP/IP **Network Access** layer.
  - **Internet Layer** (TCP/IP) = Routing messages (e.g., **IP, ICMP**).
  - **Application Layer** (TCP/IP) = **DNS, DHCP, FTP**.

🏢 Cisco Device Essentials

🧠 Memory

- **RAM** 🏃: Volatile (lost on reboot). Stores the **running-config**.
- **NVRAM** 💾: Non-volatile. Stores the **startup-config**.
- **Flash** ⚡: Stores the IOS (Operating System).

💻 Commands & Configs

- `copy running-config startup-config` 💾: Saves current changes to NVRAM. They will load on restart.
- `reload` 🔄: Restarts device. *If unsaved changes are in RAM, this discards them and loads NVRAM.*
- `show ip interface brief` 📋: Verifies switch interface status (up/down), protocol status, and IP addresses.
- `exit` 🚪: Returns to the previous level in the command hierarchy.

🔐 Security & Passwords

- `service password-encryption` 🔒: Encrypts clear-text passwords in the **configuration file stored in NVRAM** (uses weak Type 7).
  - *Note: It does NOT encrypt passwords sent over WAN links, nor does it provide strong encryption for* `enable secret`.
- **SSH vs. Telnet** 🛡️: SSH encrypts messages and uses authentication (Secure). Telnet sends everything in plaintext (Insecure).
- **Hostname Rules** 🏷️: Must start with a letter, contain only letters/numbers/hyphens, no spaces, and end with a letter/number (e.g., **RM-3-Switch-2A4**, **SwBranch799**).

🔌 Switching & Addressing

- **Default Gateway** 🚪: Routers determine the best path. On a Layer 2 switch, a default gateway is only used to allow **remote management** (SVI) traffic to leave the local subnet.
- **SVI (Switch Virtual Interface)** 👻:
  - Not a physical interface.
  - Associated with VLAN 1 by default.
  - Used for **remote management** (admin connecting from a PC/software).

- **MAC vs IP Addressing** 📬:
  - **IP Address**: End-to-end delivery (Source IP to Destination IP).
  - **MAC Address**: Hop-to-hop delivery. If the destination is on a different network, the frame's destination MAC is the **Router's MAC** (the default gateway).

🌍 WAN Internet Connection Types

- ☎️ **Dialup**: Typically has very **low bandwidth**.
- 📺 **Cable**: Uses **coaxial cable** as a medium.
- 🛰️ **Satellite**: Not suited for heavily wooded areas.
- ⚡ **DSL**: High bandwidth connection that runs over a **telephone line**.